Adaptive Tri-View 3D Pose Fusion with Cross-View Attention and Temporal Refinement

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Abstract. Robust 3D human pose estimation is essential for driver monitoring, yet monocular pose estimators often fail under occlusion, viewpoint bias, and keypoint jitter. Multi-view cameras provide complementary observations, but fusing noisy monocular 3D keypoints is non-trivial because view reliability varies over time and across body joints. We propose *TriPoseFusion*, a geometry-guided tri-view 3D pose fusion framework for recovering stable driver poses from noisy front/left/right keypoint estimates. TriPoseFusion canonicalizes poses into a body-centered coordinate frame, encodes multi-scale motion and confidence cues, performs cross-view attention with view embeddings, predicts joint-wise adaptive view gates, and refines the fused pose with a dilated temporal convolution network. The model is trained with self-supervised geometry objectives, including pseudo-teacher alignment, view consistency, bone-length regularization, temporal smoothness, contrastive view alignment, and gate entropy regularization. Experiments are designed to compare TriPoseFusion against single-view, fixed-fusion, confidence-weighted, and temporal-only baselines using MPJPE, PA-MPJPE, PCK, and robustness tests under view dropout and keypoint noise. This formulation reframes driver monitoring as a general computer vision problem: robust multi-view 3D pose fusion from noisy keypoints.

1 Introduction

Driver monitoring and in-cabin human understanding require stable 3D body pose representations. In realistic driving environments, however, monocular 3D pose estimators are frequently affected by steering-wheel occlusion, self-occlusion, viewpoint changes, low illumination, and detector noise. These factors lead to missing joints, incorrect depth estimates, and temporal jitter, which make downstream reasoning unreliable.

Multi-view camera systems alleviate some of these issues by observing the driver from complementary viewpoints. A joint invisible in the front camera may still be visible from a side camera, and pose ambiguity from one view can often be resolved by another. Nevertheless, multi-view fusion is not a simple averaging problem. The reliability of each view is time-varying and joint-dependent: one camera may be reliable for the torso but unreliable for the hands, while another may provide the opposite pattern. Fixed averaging, concatenation, or global confidence heuristics cannot model such fine-grained reliability.

We propose *TriPoseFusion*, an adaptive tri-view 3D pose fusion framework that operates on noisy monocular pose/keypoint estimates. The method first canonicalizes poses into a body-centered coordinate frame using upper-body geometry, then constructs motion-aware per-joint features from 3D positions, velocities, confidence scores, and optional reprojection cues. A cross-view attention module models complementary information among front/left/right views for each joint and frame. A joint-wise gating module predicts adaptive view weights, and a dilated temporal refiner reduces jitter by exploiting longer-range motion context.

Our contributions are summarized as follows: (1) we formulate robust driver pose recovery as noisy tri-view 3D pose fusion rather than action classification; (2) we introduce joint-wise adaptive view gating with cross-view attention to model view complementarity and reliability; (3) we design geometry-guided self-supervised objectives for training without manual 3D ground-truth annotation; and (4) we propose evaluation protocols for accuracy, temporal stability, and robustness under keypoint noise and view dropout.

2 Related Work

2.1 Monocular and Video-Based 3D Pose Estimation

Monocular 3D human pose estimation aims to recover 3D joints from a single image or video. Existing methods commonly follow either direct image-to-pose regression or a two-stage keypoint lifting paradigm, where 2D detections are first estimated and then lifted to 3D. Video-based methods further exploit temporal context to reduce frame-wise ambiguity and jitter. Despite their progress, monocular estimators remain vulnerable to occlusion, depth ambiguity, truncation, and viewpoint bias. These limitations are especially pronounced in driver monitoring, where the steering wheel, arms, and dashboard frequently occlude the upper body. Rather than designing a new monocular estimator, we treat monocular 2D/3D keypoints as noisy observations and focus on robust multi-view fusion.

2.2 Multi-View Pose Estimation and Keypoint Fusion

Multi-view pose estimation improves observability by combining evidence from multiple synchronized cameras. Classical approaches rely on camera calibration and triangulation, while recent methods fuse image features, heatmaps, or volumetric representations across views. These approaches are effective when calibrated images and reliable detections are available, but many practical systems only provide per-view pose outputs from off-the-shelf monocular estimators. In this setting, simple mean, median, or confidence-weighted fusion is often used, but such fixed rules assume that view reliability is globally consistent. TriPoseFusion instead learns joint-wise and time-varying reliability from keypoint geometry, confidence, and motion cues, allowing different joints to select different views within the same frame.

2.3 Temporal Pose Modeling and Refinement

Temporal modeling is widely used to improve pose stability. Recurrent networks, temporal convolution networks, and transformer-based models smooth short-term noise and capture motion dependencies over time. Temporal priors are particularly useful for video pose estimation because physically plausible human motion is continuous. However, temporal refinement alone cannot distinguish whether an observed artifact comes from true motion or from a corrupted camera view. Our method first performs adaptive cross-view fusion and then applies dilated temporal refinement, so temporal modeling operates on a view-aware fused pose rather than on a single noisy stream.

2.4 Self-Supervised and Geometry-Guided Pose Learning

Obtaining accurate 3D pose labels in real driving environments is difficult, motivating self-supervised and geometry-guided learning. Common supervisory signals include reprojection consistency, multi-view agreement, bone-length constraints, temporal smoothness, and pseudo labels from triangulation or robust aggregation. These constraints provide useful structure but can be brittle when some views are noisy or missing. TriPoseFusion combines pseudo-teacher alignment, view consistency, bone regularization, temporal smoothness, contrastive view alignment, and gate entropy regularization. This combination encourages geometrically plausible fused poses while preventing collapse to a single dominant view.

2.5 Positioning

In summary, prior work has made strong progress in monocular 3D pose estimation, multi-view reconstruction, and temporal refinement. However, less attention has been paid to the intermediate but practical setting where synchronized cameras provide noisy monocular keypoint estimates rather than clean images or perfectly calibrated 3D observations. TriPoseFusion targets this gap by learning adaptive, joint-wise tri-view fusion under noisy keypoints, making it suitable for real driver monitoring scenarios with occlusion and view-dependent reliability.

3 Method

3.1 Problem Formulation

Given $V = 3$ synchronized views (front/left/right), the input to TriPoseFusion is

$$\left\{P_v^{3D}, P_v^{2D}, C_v\right\}_{v=1}^{V}, \tag{1}$$

where $P_v^{3D} \in \mathbb{R}^{T\times J\times 3}$ denotes noisy monocular 3D keypoints, $P_v^{2D} \in \mathbb{R}^{T\times J\times 2}$ denotes optional 2D keypoints, and $C_v \in \mathbb{R}^{T\times J}$ denotes per-joint confidence. The output is a fused 3D pose sequence

$$P_{\text{final}} \in \mathbb{R}^{T\times J\times 3}. \tag{2}$$

The goal is to recover a temporally stable and geometrically plausible pose from noisy multi-view keypoint estimates.

3.2 Body-Centered Pose Canonicalization

Raw monocular 3D keypoints can differ in translation, orientation, and scale across views. We canonicalize each pose into a body-centered frame before fusion. Let o_t be a root joint such as neck or pelvis, and let the shoulder vector define a horizontal body axis. For each frame, we estimate a rotation R_t from the neck, left shoulder, and right shoulder, and transform each view by

$$\bar{P}_v^{t,j} = R_t \left(P_v^{t,j} - o_t \right). \tag{3}$$

When shoulder detections are unreliable, robust canonicalization rejects outlier shoulder lengths and falls back to the previous stable frame or a translation-only normalization. This step provides a geometry-aware representation that makes cross-view comparison easier.

3.3 Multi-Scale Motion-Aware View Encoding

For each view, frame, and joint, we construct a feature vector from position, motion, and reliability cues:

$$\phi_v^{t,j} = \left[\bar{P}_v^{t,j}; \dot{P}_v^{t,j}; \dot{P}_{v,k}^{t,j}; \ddot{P}_v^{t,j}; P_v^{2D,t,j}; C_v^{t,j}; e_v^{t,j} \right], \tag{4}$$

where $\dot{P}$ denotes velocity, $\dot{P}_k$ denotes multi-scale velocity over temporal stride k, $\ddot{P}$ denotes acceleration, and e_v is an optional reprojection or consistency error. A shared MLP encoder maps these features into latent tokens:

$$H_v^{t,j} = \mathrm{MLP}(\phi_v^{t,j}). \tag{5}$$

This encoding exposes both instantaneous pose and temporal dynamics to the fusion module.

3.4 Cross-View Attention with View Embeddings

For each frame t and joint j, the three views are treated as tokens. We add learnable view embeddings e_v^{view} and apply multi-head self-attention across views:

$$\tilde{H}^{t,j} = \mathrm{MHSA}\left([H_1^{t,j} + e_1^{\text{view}}, \ldots, H_V^{t,j} + e_V^{\text{view}}] \right). \tag{6}$$

This module allows the model to compare front/left/right evidence for the same joint and frame, making the fusion aware of cross-view complementarity.

3.5 Joint-Wise Adaptive View Gating

Instead of predicting a single frame-level weight, TriPoseFusion estimates joint-wise and time-varying view reliability:

$$\alpha_{t,j,v} = \mathrm{softmax}_v\left(g(\tilde{H}_v^{t,j})\right), \quad P_{\mathrm{init}}^{t,j} = \sum_{v=1}^{V} \alpha_{t,j,v} \bar{P}_v^{t,j}. \tag{7}$$

This design is important in driving scenes because different joints can be occluded in different cameras. For example, a side view may be more reliable for shoulder orientation, while the front view may better capture vertical head motion.

3.6 Dilated Temporal Pose Refinement

The initial fused pose can still contain jitter from noisy keypoints. We therefore use a dilated temporal convolution network to predict a residual correction:

$$P_{\mathrm{final}} = P_{\mathrm{init}} + \Delta P, \quad \Delta P = \mathrm{TCN}_{\mathrm{dilated}}(P_{\mathrm{init}}, \dot{P}_{\mathrm{init}}, \ddot{P}_{\mathrm{init}}). \tag{8}$$

Dilated convolutions enlarge the temporal receptive field without excessive computation, enabling the model to handle slow motion and suppress high-frequency jitter.

3.7 Training Objective

The total self-supervised objective is

$$\mathcal{L} = \lambda_{tri}\mathcal{L}_{tri} + \lambda_{view}\mathcal{L}_{view} + \lambda_{bone}\mathcal{L}_{bone} + \lambda_{temp}\mathcal{L}_{temp} + \lambda_{nce}\mathcal{L}_{nce} + \lambda_{gate}\mathcal{L}_{gate}. \tag{9}$$

$\mathcal{L}_{tri}$ aligns the fused pose to a triangulated teacher or median pseudo teacher. $\mathcal{L}_{view}$ keeps the fused pose consistent with reliable view observations. $\mathcal{L}_{bone}$ regularizes bone lengths across time, and $\mathcal{L}_{temp}$ penalizes excessive velocity or acceleration. $\mathcal{L}_{nce}$ aligns latent representations of synchronized views, while $\mathcal{L}_{gate} = \sum_{t,j,v} \alpha_{t,j,v} \log \alpha_{t,j,v}$ encourages high-entropy exploration early in training and prevents collapse to a single view.

4 Experiments

4.1 Dataset and Protocol

We evaluate on a synchronized tri-view driver monitoring dataset captured from front, left, and right cameras. A monocular pose estimator provides 2D/3D keypoints and confidence scores for each view. We split data by subject or fold to avoid identity leakage and use uniform temporal sampling for clip construction. If the dataset cannot be released publicly, we will report the capture setup, number of subjects, camera layout, and train/validation/test protocol to ensure reproducibility.

Table 1. Main pose fusion comparison. Lower MPJPE/PA-MPJPE and higher PCK are better. Values will be filled after the pose-fusion evaluation.

Method	MPJPE ↓	PA-MPJPE ↓	PCK ↑
Front only	–	–	–
Left only	–	–	–
Right only	–	–	–
Mean fusion	–	–	–
Median fusion	–	–	–
Confidence fusion	–	–	–
Temporal-only	–	–	–
TriPoseFusion	–	–	–

Table 2. Ablation design for TriPoseFusion.

Variant	Attn	TCN	Multi-scale Vel	Gate Ent.	Robust Canon	MPJPE ↓
Base	No	No	No	No	No	–
+ Dilated TCN	No	Yes	No	No	No	–
+ Multi-scale Vel	No	Yes	Yes	No	No	–
+ Gate Entropy	No	Yes	Yes	Yes	No	–
+ Robust Canon	No	Yes	Yes	Yes	Yes	–
Full	Yes	Yes	Yes	Yes	Yes	–

4.2 Evaluation Metrics

We report standard 3D pose metrics: MPJPE (mean per-joint position error), PA-MPJPE after Procrustes alignment, and PCK under distance thresholds. To measure temporal stability, we additionally report acceleration error or temporal smoothness. Robustness is evaluated by injecting Gaussian keypoint noise, masking upper-body joints, and dropping individual views at test time.

4.3 Baselines

We compare TriPoseFusion with single-view baselines (front/left/right), fixed mean fusion, median fusion, confidence-weighted fusion, a temporal-only refinement baseline, and an MLP fusion baseline. These baselines separate the effects of multi-view aggregation, confidence weighting, cross-view attention, joint-wise gating, and temporal refinement.

4.4 Main Comparison

4.5 Ablation Study

4.6 Robustness and Visualization

We perform three stress tests: dropping one camera view, adding Gaussian noise to 2D/3D keypoints, and masking upper-body joints. We expect adaptive gating to degrade gracefully compared with fixed fusion. We also visualize $\alpha_{t,j,v}$ across time, joints, and environments to show whether the model selects views consistently with visibility and occlusion patterns.

4.7 Discussion and Limitations

TriPoseFusion is designed to produce robust fused pose representations from noisy keypoints, but it still depends on the quality of the upstream pose estimator. If all three views fail simultaneously, no fusion strategy can recover missing evidence reliably. Camera synchronization and calibration errors may also affect pseudo-teacher construction and evaluation. Finally, this paper focuses on pose fusion; connecting the fused pose representation to downstream driver action recognition remains future work.

5 Conclusion

TriPoseFusion demonstrates that adaptive, geometry-guided multi-view keypoint fusion can produce more robust 3D driver pose representations than single-view or fixed-fusion baselines. By combining body-centered canonicalization, cross-view attention, joint-wise view gating, temporal refinement, and self-supervised geometry losses, the framework addresses the core challenge of noisy multi-view 3D pose fusion in realistic driver monitoring scenarios.

References